Indium Phosphide Photonics Supply Chain Factor

Capturing Value in InP Lasers, Wafers, Optical Engines & CPO
as AI Infrastructure Hits Bandwidth and Power Limits

Mathematical Framework & Implementation Specification

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Abstract

This document specifies a thematic equity factor designed to capture value shifting into Indium Phosphide (InP) light-source and photonics supply chains as AI infrastructure becomes bandwidth- and power-limited. The factor isolates companies controlling InP lasers, wafers, fabs, epitaxial processes, optical engines, and CPO-enabling capacity as 800G/1.6T/3.2T networking and silicon photonics scale across AI data centers. Scoring combines InP supply control (fabs, epi, patents), AI optical demand proof (design-wins, hyperscaler traction), capacity scarcity signals, CPO relevance, vertical integration, and strategic capital lock-in into a composite exposure score. Universe: US-listed equities in Semiconductors & Semiconductor Equipment, Electronic Equipment/Components, and Communications Equipment. Overall confidence: 0.72–0.78.

Contents

1 Executive Summary

1.1 Factor Theme

Capture value shifting into Indium Phosphide light-source and photonics supply chains as AI infrastructure becomes bandwidth- and power-limited. Isolates companies controlling:

- InP lasers (DFB, EML, CW), wafers, fabs, and epitaxial (epi) processes
- Optical engines and photonic integrated circuits (PICs)
- CPO-enabling capacity for 800G/1.6T/3.2T networking
- Silicon photonics external laser sources

1.2 Investment Rationale

AI networking is hitting bandwidth and power walls. Value migrates to optical sources and tightly integrated photonics where:

- Capacity, process know-how (epi/lasers), and customer lock-ins enable scarcity rents
- 6-inch InP wafer transition advantages create supply bottlenecks
- Hyperscaler equity investments and multiyear purchase commitments de-risk revenue visibility
- Vertical integration (in-house epi/wafer capability) enables pricing power

1.3 Target Industry Sectors

GICS Industry	Coverage
IT — Semiconductors & Equipment	InP wafer/substrate producers, epitaxial growers, laser chip fabs
IT — Electronic Equipment & Components	Optical transceivers, PICs, optical engine assemblers
IT — Communications Equipment	CPO module suppliers, networking platforms with CPO

Table 1: Target GICS industry mappings

1.4 Avoid List (Signal G)

- Generic optical component names with weak InP exposure
- Early-stage photonics stories without production evidence
- Companies where telecom/legacy optics dominates AI upside
- Passive fiber/connectors without InP/laser/epi exposure

2 Signal Taxonomy (A–G)

2.1 Signal A: InP Supply Control

Definition: Exposure to InP fabs, InP wafers, epiwafers, DFB/EML/CW lasers, photodiodes, or 6-inch InP scaling.

Sources: Segment mix, fab disclosures, capex announcements, patent filings.

Key Evidence:

- Ownership of InP-relevant fabs or wafer lines

- In-house epi/wafer/epitaxy processes
- 6-inch InP capability flag
- Patents in InP lasers, DFB/EML/CW, InP epi/heterostructures
- Facility descriptors in SEC filings (capex, wafer size transitions)

2.2 Signal B: AI Optical Demand Proof

Definition: Revenue or commentary tied to 800G, 1.6T, 3.2T, CPO, silicon photonics, external laser sources, or hyperscaler optical ramps.

Sources: Earnings transcripts, design-win mentions, investor presentations.

Key Evidence:

- Explicit 800G/1.6T/3.2T revenue or design-wins
- CPO/silicon photonics customer qualifications
- Hyperscaler customer mentions (AWS, Azure, GCP, Meta)
- Press releases confirming strategic partnerships

2.3 Signal C: Capacity Scarcity

Definition: Evidence of sold-out orders, capacity expansion, new fabs, yield constraints, purchase commitments, or customers securing future supply.

Sources: Backlog duration, lead-time disclosures, 8-K announcements.

Key Evidence:

- "Sold out", "capacity constrained" language
- Backlog months or lead-time weeks disclosed
- Announced capacity expansions or new fab builds
- Multiyear supply/commitment agreements

2.4 Signal D: CPO Relevance

Definition: Products used in external laser sources, optical engines, photonic integrated circuits, wafer-level optical test, or CPO packaging.

Sources: Product announcements, customer qualifications, patent filings.

Key Evidence:

- External laser source products
- Optical engine or PIC manufacturing
- Wafer-level optical test capability
- CPO module integration or qualification

2.5 Signal E: Pricing Power & Vertical Integration

Definition: Ability to capture scarcity rents through EML/CW price increases, in-house epi/wafer capability, or 6-inch wafer transition advantages.

Sources: ASP trends, gross margin trajectory, vertical integration disclosures.

Key Evidence:

- Price increase announcements for EML/CW lasers
- Gross margin expansion from product mix
- In-house vs. external supplier reliance

- 6-inch wafer transition advantages

2.6 Signal F: Strategic Capital Lock-In

Definition: Hyperscaler equity investments, multibillion-dollar purchase commitments, or multiyear capacity reservations.

Sources: Partnership announcements, commitment duration/value.

Key Evidence:

- NVIDIA/hyperscaler strategic partnerships (e.g., Lumentum-NVIDIA, Coherent-NVIDIA)
- Equity investments or prepayments from customers
- Long-term agreements (LTAs) with capacity reservations
- Multiyear commitment disclosures in filings

2.7 Signal G: Avoid Criteria (Negative)

Definition: Penalty for legacy telecom dominance or weak InP/AI optical exposure.

Triggers:

- Legacy telecom optics majority of revenue
- Generic optical components without InP exposure
- Low topic relevance scores for InP/CPO/800G
- Pre-revenue photonics without production evidence

3 Exposure Scoring Dimensions

3.1 Dimension D1: InP Supply Control ($w = 0.35$)

Definition: Asset and technology ownership in InP fabs, wafers, epi, and lasers.

Measurement Fields:

- Number/type of InP-relevant fabs or owned wafer lines
- 6-inch InP capability flag (binary)
- In-house epi/wafers/epitaxy processes (% of production)
- Patent counts: InP lasers, DFB/EML/CW, epi/heterostructures

Evidence Sources: SEC filings (10-K/10-Q/8-K), investor presentations, patents_adapter.

3.2 Dimension D2: AI Optical Demand Proof ($w = 0.25$)

Definition: Design-wins and hyperscaler traction for 800G+ / CPO / silicon photonics.

Measurement Fields:

- Explicit revenue/design-win mentions for 800G/1.6T/3.2T/CPO
- Earnings-transcript mention frequency for target keywords
- Press releases confirming hyperscaler partnerships

Evidence Sources: fmp_adapter.get_earnings_transcripts, investor presentations, press releases.

3.3 Dimension D3: Capacity Scarcity & Lock-In ($w = 0.20$)

Definition: Evidence of sold-out capacity, backlog, lead-times, and customer commitments.

Measurement Fields:

- Statements of sold-out capacity or backlog length
- Announced capacity expansions or new fabs
- Customer commitments or equity investments
- Contract value/term disclosures

Evidence Sources: SEC 8-K, earnings transcripts, press releases.

3.4 Dimension D4: Vertical Integration & Pricing Power ($w = 0.20$)

Definition: Gross margin trajectory, in-house capability, and pricing power evidence.

Measurement Fields:

- Gross margin trend from financials
- In-house wafer/epi vs. external supplier reliance
- Price increase announcements for EML/CW
- Premium ASP evidence

Evidence Sources: SEC filings, fmp_adapter, transcript pricing mentions.

4 Composite Score Construction

4.1 Exposure Scoring Composite Formula

$$\boxed{\text{Score} = 0.35 \cdot S_{D1} + 0.25 \cdot S_{D2} + 0.20 \cdot S_{D3} + 0.20 \cdot S_{D4}} \tag{1}$$

Each S_{D*} is scaled to $[0, 1]$ via robust min–max normalization after winsorizing at 5th/95th percentiles.

4.2 Factor Methodology Composite Formula (7 Dimensions)

$$\text{Score} = \max(0,\ 0.30 \cdot Z(D1) + 0.20 \cdot Z(D2) + 0.15 \cdot Z(D3) + 0.15 \cdot Z(D4) + 0.10 \cdot Z(D5) + 0.10 \cdot Z(D6) - 0. \tag{2}$$

4.3 Scoring Dimension Formulas

4.4 Normalization Method

1. Per-dimension robust cross-sectional min–max after winsorizing at 5th/95th percentiles
2. Fallback: Compute z-score across universe, convert to percentile via Gaussian CDF
3. Binary indicators mapped {false, true} → {0, 1} before winsorization
4. Final composite scaled to $[0, 1]$ via min–max

4.5 Score Interpretation

High Score: Company uniquely controls InP-relevant assets (fabs/epi/6-inch capability or in-house lasers) AND has verified AI optical demand (800G+/CPO/hyperscaler design-wins) with evidence of capacity scarcity or binding commitments—implying material ability to capture scarcity rents.

Dimension	Formula	Weight
D1: InP Supply Control	$0.5 \cdot \text{rank}(\log(1+\text{patents}_{5y})) + 0.3 \cdot \text{epi\_flag} + 0.2 \cdot \text{6in\_flag}$	0.30
D2: AI Demand Proof	$0.7 \cdot \text{rank}(\text{hits\_ai\_demand}) + 0.3 \cdot \max(\text{designwin}, \text{hyperscaler})$	0.20
D3: Capacity Scarcity	$0.25 \cdot \text{soldout} + 0.25 \cdot \min(\text{backlog}/12, 1) + 0.20 \cdot \min(\text{leadtime}/26, 1) + 0.15 \cdot \text{capex\_fab} + 0.15 \cdot \text{lta}$	0.15
D4: CPO Relevance	$0.6 \cdot \text{mean}(\text{cpo\_flags}) + 0.4 \cdot \text{rank}(\log(1 + \text{patents\_cpo}_{5y}))$	0.15
D5: Pricing Power	$0.5 \cdot \text{mean}(\text{vertical\_flags}) + 0.3 \cdot \text{rank}(\text{momentum\_quality}) + 0.2 \cdot \text{price\_increase}$	0.10
D6: Strategic Lock-In	$0.5 \cdot \text{lta} + 0.25 \cdot \text{capacity\_reservation} + 0.25 \cdot \text{equity\_prepay}$	0.10
D7: Avoid Penalty	$0.5 \cdot \text{legacy\_telecom} + 0.5 \cdot (1 - \text{topic\_relevance})$	-0.20

Table 2: Scoring dimension formulas

Low Score: Weak or indirect InP exposure (component reseller or telecom legacy), no design-wins nor capacity constraints, or early-stage R&D stories lacking production evidence.

5 Portfolio Weight Construction

5.1 Base Weight Formula

$$w_i = \frac{(\text{FinalScore}_i)^{1.0} \times (\text{ADV}_{60,i})^{0.5}}{\sum_j (\text{FinalScore}_j)^{1.0} \times (\text{ADV}_{60,j})^{0.5}} \tag{3}$$

Liquidity-adjusted weighting ensures tradability while preserving score-proportionality.

5.2 Constraints and Caps

Constraint	Value
Single-Name Maximum	7%
Single-Name Minimum	0.3%
Sub-Industry Cap	40%
Top-5 Aggregate Cap	35%
Maximum Quarterly Turnover	80%

Table 3: Portfolio weight constraints

5.3 Constraint Application Order

1. Apply sub-industry cap (40%); redistribute excess pro-rata
2. Apply single-name caps (0.3%–7%); redistribute excess
3. Apply top-5 aggregate cap (35%); redistribute if exceeded
4. Renormalize to sum to 100%

6 Tradability Filters

Filter	Threshold	Source
Minimum 60-day ADV	$\geq$ $3,000,000	polygon_adapter.avg_dollar
Minimum Price	$\geq$ $5.00	polygon_adapter.get_daily_
Minimum Market Cap	$\geq$ $300,000,000	Market data
Minimum Free Float	$\geq 10\%$	Market data
Maximum Amihud Illiquidity	$\leq 5 \times 10^{-5}$	Computed from OHLCV
Listing	US primary listing only	Exchange data
Excluded	OTC, Pink Sheets, delisted, special situations	Listing status

Table 4: Tradability filter thresholds

7 Classification Tiers

Tier	Score Range	Weight Range	Characteristics
Tier 1: Direct InP Core	$[0.75, 1.0]$	$[3\%, 7\%]$	InP fabs, epi, laser chips, wafer ownership
Tier 2: Optical Engine/CPO	$[0.50, 0.75)$	$[1\%, 4\%]$	CPO modules, optical engines, PICs
Tier 3: Peripheral	$[0.25, 0.50)$	$[0.3\%, 2\%]$	Indirect exposure, transceiver assembly

Table 5: Classification tiers by composite score

8 Data Requirements

8.1 Adapter Data Requirements

8.1.1 sec_adapter (MUST HAVE)

- `get_filings`: 10-K, 10-Q, 8-K, S-1, 424B5 for 3-year lookback
- Fields: segment descriptions, manufacturing/capacity disclosures, major customers, exhibit links
- Use cases: Facility, capex, backlog, formal contract disclosures

8.1.2 fmp_adapter (MUST HAVE)

- `get_earnings_transcripts`: Quarterly transcripts for 5-year lookback
- Fields: transcript text, revenue, R&D expense
- Use cases: 800G/1.6T/3.2T/CPO mentions, design-wins, backlog commentary

8.1.3 patents_adapter (MUST HAVE)

- `get_patents_by_tickers_batch`: Patent records by CPC/IPC
- CPC codes: InP lasers, DFB/EML/CW, epitaxial processes, photodiodes, CPO packaging
- Use cases: Technology ownership, patent intensity, recent grants

8.1.4 company_overview (MUST HAVE)

- `get_company_profile`: Industry, sector, description, CIK
- `check_topic_relevance`: InP, DFB, EML, CW laser, 800G, CPO, silicon photonics

8.1.5 polygon_adapter (MUST HAVE)

- `get_daily_close_prices`: Price history for returns
- `avg_dollar_volume`: 60-day ADV for liquidity
- `get_ohlcv`: For Amihud illiquidity and momentum calculations
- `get_short_interest`: Days-to-cover

8.2 External Sources

Domain	Data Types
mofcom.gov.cn	Export controls for indium-related materials (supply risk)
semi.org (M23 spec)	InP wafer specifications, 6-inch transition parameters
oiforum.com	OIF CMIS and module interface standards
snia.org	SFF module management interface docs
coherent.com	Investor presentations, partnership announcements
investor.lumentum.com	Press releases, NVIDIA partnership
globenewswire.com	Order announcements (e.g., AOI 800G)
lightcounting.com	Industry market intelligence

Table 6: External data sources

8.3 Missing Data Requirements

1. Machine-readable capacity metrics (InP wafers/month, epi throughput) — typically qualitative only
2. Product-level ASPs (EML/CW laser price time series) — commercial, often not disclosed
3. Automated investor-deck parsing adapter — requires web scraping

9 Keyword Dictionaries

9.1 InP / Laser Keywords

- InP, indium phosphide, InP wafer, InP substrate, 6-inch InP, 150mm InP
- DFB laser, EML, CW laser, continuous wave, distributed feedback
- Epitaxy, epi, epiwafer, MOCVD, MBE, heterostructure
- Photodiode, avalanche photodiode, InGaAs

9.2 AI Optical Demand Keywords

- 800G, 1.6T, 3.2T, 400G (context-dependent)
- CPO, co-packaged optics, silicon photonics, SiPh
- External laser, optical engine, PIC, photonic integrated circuit
- Hyperscaler, AI networking, AI fabric, GPU interconnect

9.3 Capacity / Scarcity Keywords

- Sold out, capacity constrained, fully utilized
- Backlog, lead time, allocation, prioritized customers
- Capacity expansion, new fab, fab build, capex

- Long-term agreement, LTA, purchase commitment, prepayment

9.4 Legacy / Avoid Keywords

- Telecom, DWDM, metro, long-haul (without AI context)
- Legacy optics, traditional transceiver
- Passive fiber, fiber connector, generic component

10 Execution Pipeline

10.1 Phase 1: Universe Construction (Steps 1–5)

Step	Name	Agent	Description
1	universe_selection	UniverseSelectionAgent	S&P 500 + Russell 2000, GICS filter
2	sec_filings	GenericExtractionAgent	10-K/10-Q/8-K for 3 years
3	earnings_transcripts	GenericExtractionAgent	Quarterly transcripts (5 years)
4	patents_extraction	GenericExtractionAgent	Patents by CPC (10 years)
5	company_profiles	GenericExtractionAgent	Profiles + topic relevance

Table 7: Phase 1 execution steps

10.2 Phase 2: Data Collection & Metrics (Steps 6–10)

Step	Name	Agent	Description
6	prices_liquidity	GenericExtractionAgent	OHLCV, ADV, short interest
7	text_classification	ClassificationAgent	Label signals A–F from text
8	derive_features	GenericExtractionAgent	Patent counts, text metrics
9	llm_scoring	LLMScoringAgent	Per-dimension scores D1–D4
10	price_metrics	CodeGeneratingAgent	Momentum, Amihud, ADV

Table 8: Phase 2 execution steps

10.3 Phase 3: Scoring & Portfolio (Steps 11–15)

Step	Name	Agent	Description
11	dimension_scoring	GenericScoringAgent	Apply formulas D1–D7
12	composite_score	CodeGeneratingAgent	Compute weighted composite
13	tradability_filter	CodeGeneratingAgent	ADV/price/Amihud filters
14	rank_and_weight	CodeGeneratingAgent	Caps, liquidity adjustment
15	generate_outputs	OutputAgent	Loadings, audit trail, reports

Table 9: Phase 3 execution steps

11 Rebalancing Rules

11.1 Schedule

- **Frequency**: Quarterly on third Friday of Mar/Jun/Sep/Dec
- **Effective**: Next trading day open
- **Data Cutoff**: T−5 trading days before effective date

11.2 Buffer Rules

- **Entry Buffer**: New entrants require rank $\geq$ 60th percentile
- **Exit Buffer**: Existing members remain if rank $\geq$ 50th percentile
- **Purpose**: Reduce turnover from borderline names

11.3 Turnover Guardrails

- Maximum quarterly turnover: 80%
- If projected turnover >50%: Use VWAP over 2–3 days
- Off-cycle: Remove delisted names at next close; renormalize

12 Risk and Validation

12.1 Validation Criteria

1. **Sanity**: >80% of constituents have non-zero signals in $\geq$3 positive dimensions
2. **Backtest**: At least 36 months of simulated quarterly rebalances
3. **Event Validation**: D2/D3 score increases cluster around earnings mentioning 800G/CPO
4. **Liquidity**: Projected participation $\leq$10% of ADV per name
5. **Stability**: 1-month score autocorrelation $\geq$0.7 for D1/D4 patent/fab signals

12.2 Constraints Summary

- Liquidity: $\mathrm{ADV}_{60d} \geq \$3\mathrm{M}$
- Price: Last close $\geq$ $5.00
- Amihud illiquidity: $\leq 5 \times 10^{-5}$ over 60 days
- No OTC/Pink; US primary listings only
- GICS sub-industry cap 40%; single-name cap 7%; top-5 aggregate $\leq$35%

12.3 Key Limitations

1. No adapter for automated parsing of investor decks or press releases
2. Product-level capacity and ASP time-series not disclosed in structured filings
3. Some contract values/terms may be redacted or qualitative only
4. Text-based signals may lag true operational constraints
5. Patent counts reflect R&D direction, not guaranteed commercialization

13 Sample Companies

Ticker	Name	Role / Relevance
COHR	Coherent Corp	InP lasers, optical engines, NVIDIA partnership
LITE	Lumentum	InP lasers, EML/CW, NVIDIA partnership
AAOI	Applied Optoelectronics	800G transceivers, hyperscaler orders
IIVI	II-VI (now Coherent)	InP epi, vertical integration
–	IQE plc	InP epi/wafer supplier

Table 10: Sample companies (illustrative; actual inclusion via scoring pipeline)

14 Citations

14.1 Industry & Standards Sources

1. MOFCOM Export Control Announcement (Feb 4, 2025) — Indium-related export controls. https://www.mofcom.gov.cn/zcfb/blgg/art/2025/art_b31fd5c823484faea7595290eed12acc.html
2. SEMI M23 Specification — Polished monocrystalline InP wafers. https://store-us.semi.org/products/m02300-semi-m23-specification-for-polished-monocrystalline-indium-pho
3. OIF C-CMIS Specification — Module interfaces for CPO. https://www.oiforum.com/wp-content/uploads/OIF-C-CMIS-01.4.pdf
4. ScienceDirect — Enabling MOCVD for 150mm (6-inch) InP wafers. https://www.sciencedirect.com/science/article/pii/S0022024824002288

14.2 Company-Specific Sources

1. Lumentum: NVIDIA Strategic Partnership Announcement. https://investor.lumentum.com/financial-news-releases/news-details/2026/NVIDIA-Announces-Strategic-Partnership-Wi default.aspx
2. Coherent: NVIDIA Partnership and OFC 2026 Investor Deck. https://www.coherent.com/news/press-releases/nvidia-and-coherent-announce-strategic-partnership
3. AOI: 800G Transceiver Upsized Order (Apr 2026). https://www.globenewswire.com/news-release/2026/04/02/3267838/0/en/AOI-Receives-New-Upsized-Order-for-800G-Data-Cente html
4. IQE: InP Product Pages. https://www.iqep.com/products/inp/

14.3 Index Methodology References

1. FTSE Russell Capping Methodology Guide (v5.0, Jan 2026). https://research.ftserussell.com/products/downloads/Capping_Methodology_Guide.pdf
2. S&P Index Mathematics Methodology. https://www.spglobal.com/spdji/en/documents/methodologies/methodology-index-math.pdf
3. MSCI Thematic Relevance Score Methodology (May 2022). https://www.msci.com/indexes/documents/methodology/2_MSCI_Thematic_Relevance_Score_Methodology_20220519.pdf
4. Russell US Indexes Construction Methodology. https://www.lseg.com/content/dam/ftse-russell/en_us/documents/ground-rules/russell-us-indexes-construction-and-methodolo pdf

A Appendix A: Extraction Metadata

B Appendix B: CPC Code Mappings

C Appendix C: Step-by-Step Implementation Guide

1. **Define Universe**: UniverseSelectionAgent loads S&P 500 + Russell 2000; apply GICS filter for IT (Semiconductors, Electronic Equipment, Communications Equipment).
2. **Firm Profiles & Topic Gate**: company_overview.get_company_profile + check_topic_relevance for "InP", "optical transceiver", "CPO", "silicon photonics".
3. **SEC Filings**: sec_adapter.get_filings for 10-K/10-Q/8-K over 24 months.
4. **Earnings Transcripts**: fmp_adapter.get_earnings_transcripts for last 5 quarters.
5. **Patents**: patents_adapter.get_patents_by_tickers_batch with CPC inference for InP/lasers/epi/photo

Field	Value
Extraction Timestamp	2026-05-01T16:18:29.275561+00:00
Pipeline Version	4.0.0
Documents Processed	198
Total Snippets Extracted	304
High-Confidence Threshold	0.5
Snippets After Filtering	90
Classification Model	gpt-5-mini
Agent Extraction Model	gpt-5-mini
Overall Confidence (Universe/Data)	0.72
Overall Confidence (Exposure Scoring)	0.78
Overall Confidence (Factor Methodology)	0.78

Table 11: Extraction metadata

Technology	Relevant CPC Codes
InP Lasers / DFB / EML	H01S 5/00, H01S 5/10, H01S 5/323
Epitaxial Processes	C30B 25/00, C30B 29/40, H01L 21/02
Photodiodes	H01L 31/10, H01L 31/105
III-V Semiconductors	H01L 29/20, H01L 33/00
Optical Modules / CPO	G02B 6/42, H04B 10/00
Wafer-Level Test	G01R 31/26, H01L 22/00
Silicon Photonics	G02B 6/12, H01P 3/00

Table 12: CPC code mappings for InP/optics patent filtering

6. **Market Data**: polygon_adapter for OHLCV (252 days), ADV (60 days).
7. **Classify Disclosures**: ClassificationAgent extracts: inhouse_epi_flag, inp_wafer_flag, laser_fab_flag, 6in_flag, keyword hits (800G/1.6T/3.2T/CPO), designwin_flag, soldout_flag, backlog_months, leadtime_weeks, lta_flag.
8. **Text Metrics**: CodeGeneratingAgent computes recency-weighted keyword frequencies (half-life 6 months), normalize per 1,000 words.
9. **Patent Metrics**: Compute per-ticker counts in CPC groups; log-scale: $\log(1 + \text{count})$.
10. **Price Proxies**: 6-month momentum, realized volatility, momentum_quality $= \text{ret}_{126}/\text{vol}_{126}$, Amihud illiquidity.
11. **Dimension Scoring**: GenericScoringAgent applies D1–D7 formulas; winsorize at 1st/99th pct; z-score cross-sectionally.
12. **Composite Score**: Compute weighted sum; apply avoid penalty; ensure non-negative.
13. **Tradability Filters**: ADV $\geq$ \$3M, price $\geq$ \$5, Amihud $\leq 5 \times 10^{-5}$.
14. **Ranking & Weighting**: $\text{Score}^{1} \times \text{ADV}^{0.5}$; apply sub-industry 40%, single-name 7%, top-5 35% caps.
15. **Output**: Factor loadings, selection audit (flags/quotes), turnover, Excel/HTML rebalance pack.